J. Earth Syst. Sci. (2018) 127:3
https://doi.org/10.1007/s12040-017-0910-3

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New paleomagnetic results on ∼2367 Ma Dharwar giant dyke swarm, Dharwar craton, southern India: implications for Paleoproterozoic continental reconstruction

N Ramesh Babu[1,*], M Venkateshwarlu[1], Ravi Shankar[1], E Nagaraju[2] and V Parashuramulu[1]

[1] *CSIR-National Geophysical Research Institute, Uppal Road, Hyderabad 500 007, India.*
[2] *Centre for Earth and Space Sciences, University of Hyderabad, Hyderabad 500 046, India.*
**Corresponding author. e-mail: nrbabungri@gmail.com*

MS received 12 January 2017; revised 26 June 2017; accepted 30 June 2017; published online 14 February 2018

Here we report new paleomagnetic results and precise paleopole position of the extensional study on ∼2367 Ma mafic giant radiating dyke swarm in the Dharwar craton, southern India. We have sampled 29 sites on 12 dykes from NE–SW Karimnagar–Hyderabad dykes and Dhone–Gooty sector dykes, eastern Dharwar craton to provide unambiguous paleomagnetism evidence on the spectacular radiating dyke swarm and thereby strengthening the presence of single magmatic event at ∼2367 Ma. A total of 158 samples were subjected to detailed alternating field and thermal demagnetization techniques and the results are presented here along with previously reported data on the same dyke swarm. The remanent magnetic directions are showing two components, viz., seven sites representing four dykes show component (A) with mean declination of 94° and mean inclination of − 70° (k = 87, α_{95} = 10°) and corresponding paleopole at 16°N, 41°E (dp = 15° and dm = 17°) and 22 sites representing 8 dykes yielded a component (B) with mean declination of 41° and mean inclination of − 21° (k = 41, α_{95} = 9°) with a paleopole at 41°N, 200°E (dp = 5° and dm = 10°). Component (A) results are similar to the previously reported directions from the ∼2367 Ma dyke swarm, which have been confirmed fairly reliably to be of primary origin. The component (B) directions appear to be strongly overprinted by the 2080 Ma event. The grand mean for the primary component (A) combined with earlier reported studies gives mean declination of 97° and mean inclination of − 79° (k = 55, α_{95} = 3°) with a paleopole at 15°N, 57°E (dp = 5°, dm = 6°). Paleogeographical position for the Dharwar craton at ∼2367 Ma suggests that there may be a chance to possible spatial link between Dharwar dykes of Dharwar craton (India), Widgemooltha and Erayinia dykes of Yilgarn craton (Australia), Sebanga Poort Dykes of Zimbabwe craton (Africa) and Karelian dykes of Kola-Karelia craton (Baltica Shield).

Keywords. Paleomagnetism; Dharwar giant dyke swarm; Dharwar craton; India.

1. Introduction

Paleomagnetic, rock magnetic and associated studies on mafic dykes are key components in the geodynamic interpretation of past continents. Mafic dykes are wide spread on the Indian shield, and range in age from Archaean to Cretaceous (Kumar *et al.* 2001; Srivastava 2010). Approximately 140,000 km^2 area is covered by a 2367 Ma radiating dyke swarm in the Dharwar craton, known

as the Dharwar giant dyke swarm (DGDS) (Halls *et al.* 2007; Kumar *et al.* 2012). The trends of the radiating dykes range between N50°E and N120°E and the fan angle is about 70°. Convergence is positioned at a focal point about 300 km west of the present day Dharwar craton boundary (French and Heaman 2010; Kumar *et al.* 2012). A total of nine U–Pb and Pb–Pb ages have been reported (Halls *et al.* 2007; French and Heaman 2010; Demirer 2012; Kumar *et al.* 2012) and these ages range between 2365–2369 Ma on this dyke swarm. Therefore, this dyke swarm was emplaced geologically in a very short time span. Several workers have previously reported rock magnetic, paleomagnetic, geochemical and geochronology results on many dykes from the DGDS (Radhakrishna and Joseph 1996; Piispa *et al.* 2011; Dash *et al.* 2013; Belica *et al.* 2014). Paleomagnetism and ages have been reported at northern and southern part of the DGDS but not from central part of the ∼ 2367 Ma swarm which forms the focus of our study. The main purpose of this study is to further establish the geographic span of the ∼ 2367 Ma dyke swarm and also to identify the nature and areal extent of the secondary magnetization. We have also attempted a paleogeographic reconstruction using well dated paleomagnetic poles for coeval mafic dyke events at ca. 2.4 Ga from different cratons.

2. Geology and sampling

Archaean basement of Dharwar craton occupies 350,000 km^2 and cross cut by most of the Precambrian dyke swarms, sills and mafic–ultramafic intrusions in southern peninsular India (Ernst and Srivastava 2008; Meert and Pandit 2015). The Dharwar craton is divided into two regions with distinct geology in the western (3.0–3.3 Ga) and eastern (2.7–2.9 Ga) Dharwar cratons, separated by the Chitradurga thrust (Chadwick *et al.* 2000 and references therein). Amalgamation of the west and east Dharwar segments is considered to have taken place at 2.5 Ga (Balakrishnan *et al.* 1991; Chadwick *et al.* 2000; Meert *et al.* 2010) and is marked by the N–S trending Closepet Granite. The extensive supracrustal greenstones are broad and flat in the west but form narrow and arcuate belts with granitic gneisses to the east (Radhakrishna and Ramakrishna 1990). The Dharwar craton is bounded by the Eastern Ghats Mobile Belt (EGMB) in east and to the south by the Southern Granulite Terrane (SGT). The Godavari Rift defines the NE margin; the Central Indian Tectonic Zone (CITZ) marks northern boundary (Rogers 1986) and in the west it is truncated by the Arabian Sea (Aggarwal *et al.* 1992). Mafic dyke swarms are widespread in the Dharwar craton with their densest concentration in eastern and lesser at western Dharwar craton (figure 1) with variably trending dykes (NE–SW, E–W, N–S and NW–SE). Dolerite dykes from Suryapet, Hyderabad and Dhone–Gooty areas are coarse-grained with massive texture and black to greyish colour and exhibit similar NE–SW trends. These dykes vary in length from few meters to hundreds of kilometers and their width vary between 1 and ∼ 300 m. These dykes range in age from Early Proterozoic (Pandey *et al.* 1997) to Late Cretaceous (Kumar *et al.* 2001). Amongst these different mafic dyke swarms, the ∼ 2367 Ma radiating mafic dyke swarm is the most dominant (Halls *et al.* 2007; French and Heaman 2010; Kumar *et al.* 2012) and oldest in Dharwar craton. There are no reported coeval magmatic events at 2367 Ma elsewhere in India.

A total of 158 oriented block samples were collected (oriented with magnetic compass) at 29 sites on 12 dykes from seven traverses on NE–SW trending dykes. Paleomagnetic sampling covers an area of Hyderabad and Karimnagar town (24 sites) and Dhone–Gooty town (five sites) surroundings on the DGDS in the vicinity of Cuddapah Basin (figure 1). Most of the samples collected are from dyke margins, fresh and in-situ road cuttings, stream cuttings and quarries.

3. Paleomagnetic investigations

All the collected oriented samples were analysed for paleomagnetic investigations and subjected to standard demagnetization procedures including incremental alternating field (AF) and thermal treatment (TH) using Molspin Alternating Field demagnetizer and MMTD 80 oven (Magnetic Measurements, UK) after measuring their initial natural remanent magnetization (NRM) using either Molspin (Magnetic Measurements, UK) or JR-6 spinner magnetometer (Agico, Czechoslovakia), respectively. AF demagnetization was performed on each sample in successive steps at closely spaced intervals from 2.5 to 20 mT and then at 10 mT intervals until a stable end point

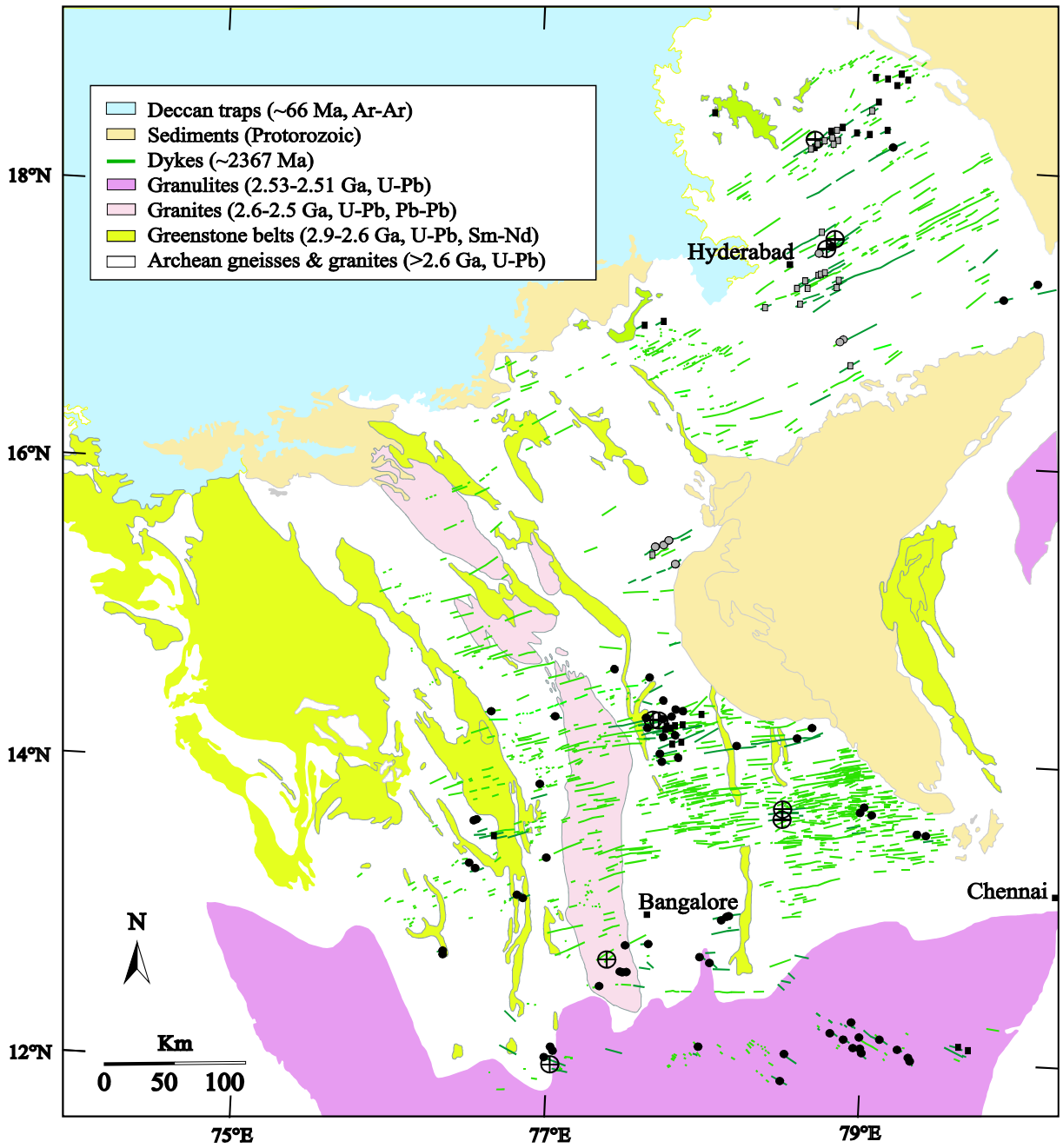


Figure 1. Simplified geological map of Dharwar craton (modified after Geological Survey of India 1998 and Naqvi and Rogers 1987) showing the distribution of ca. 2367 Ma dykes (dyke distribution after Kumar *et al.* 2012, French and Heaman 2010, Halls *et al.* 2007 and Google Earth maps) and locations of sampled sites. Open circles (component A) represent present study, solid circles represent published work and open squares (component B) represent present study and solid squares represent published work (references in table 1). U–Pb/Pb–Pb baddeleyite geochronology studies were performed on sites represented by cross in circle, ages reported by Halls *et al.* (2007); French and Heaman (2010); Demirer (2012); Kumar *et al.* (2012). Sample numbers are given in table 1.

was established. Step-wise thermal demagnetization was performed in progressive steps from 50° to a maximum of 600°C. The directional data were analyzed using Principal Component Analysis (PCA) (Kirschvink 1980) in combination with Zijderveld diagrams (Zijderveld 1967) as per standard paleomagnetic procedures. Results are shown in tables 1 and 2 and plotted in figures 2 and 3. Isothermal remanent magnetization (IRM) studies were conducted using Pulse Magnetiser (MMPM 10) and curves are shown in figure 4.

Table 1. *Summary of paleomagnetic results from ~2367 Ma dykes with component (A).*

Site	Lat (°N)	Long (°E)	N/n	D (°)	I (°)	k	α_{95} (°)	Plat. (°N)	Plong. (°E)	dp (°)	dm (°)	Ref
Dyke 1												This study
HY 4	17.423	78.718	14	128	−68	52	6	37	40	8	9	
Dyke 2												
DK1*	16.815	78.844	11	67	−64	27	9	3	39	11	14	”
DK2*	16.816	78.845	11	114	−69	163	4	28	40	5	6	”
Mean	16.815	78.844	22	88	−68	29	6	12	39	8	10	”
Dyke 3												
DG2*	15.551	77.783	11	56	−72	124	4	−5	51	6	7	”
DG3*	15.564	77.805	11	68	−71	100	5	1	45	7	8	”
DG4*	15.560	77.841	8	87	−67	71	7	10	36	9	11	”
Mean	15.558	77.809	30	69	−70	69	3	1	45	5	6	
Dyke 4												
DG17*	15.348	77.768	9	89	−69	37	8	12	40	12	15	”
4 dykes mean	***16.904***	***78.481***	***7/75***	***94***	***−70***	***87***	***10***	***16***	***41***	***15***	***17***	”
5 Sites												***1 and 2***
Site14*	14.140	77.630	10	83	−77	44	4	10	53	6	7	1
Site16*	14.170	77.740	14	22	−84	31	5	3	73	10	10	1
71*^	14.196	77.810	6	75	−72	325	4	4	46	6	7	2
596*^	14.192	77.796	7	85	−80	182	5	12	59	9	10	2
BU*^	14.198	77.808	12	91	−75	71	5	13	49	9	9	2
Mean	**14.179**	**77.756**	**5/49**	**78**	**−78**	**165**	**6**	**9**	**57**	**11**	**12**	
2	13.290	76.463	4	22	−76	43	14	−12	67	24	26	2
16*	13.183	77.041	8	339	−81	78	6	−3	83	12	12	2
28*	13.334	79.405	6	256	70	95	7	3	44	10	12	2
41*	13.540	79.005	6	117	−79	176	5	22	59	9	10	2
45*	13.533	79.016	5	331	78	131	7	33	66	12	13	2
62*	14.156	78.151	7	74	−85	80	7	12	68	13	13	2
590	14.474	77.626	4	17	−76	90	10	−11	70	17	18	2
592*	14.313	77.637	6	234	86	42	9	10	71	17	17	2
5118	13.255	76.449	7	12	−81	17	16	−4	73	30	31	2
GR*	13.972	77.834	7	238	72	34	11	−4	50	17	19	2
TN^	14.388	76.920	8	116	−77	17	14	24	52	24	25	2
GT*^	14.230	77.632	6	260	82	61	9	11	62	16	17	2
P38*	14.450	77.700	6	196	77	42	10	−9	71	17	19	3
P69	14.570	77.380	6	209	78	14	19	−5	67	33	35	3
P53*	17.220	80.110	5	305	85	15	13	22	72	25	26	3
P78	17.160	79.800	4	165	−79	27	18	38	73	32	34	3
P21*	14.230	78.750	5	23	−68	79	9	−22	64	13	15	3
2 sites												***4***
P37*	12.060	79.350	5	198	−76	716	3	37	90	5	5	4
P58*	12.110	79.250	7	175	−76	100	6	39	76	10	11	4
Mean	**12.085**	**79.300**	**12**	**186**	**−76**	**429**	**12**	**38**	**83**	**21**	**22**	
P38	12.050	79.330	4	175	−79	88	10	33	77	18	19	4
P59	12.160	79.160	4	302	−75	100	9	−4	103	15	17	4
P42*	12.200	78.900	8	130	−72	129	5	31	49	8	9	4
Dyke3*	18.127	79.220	8	69	−65	20	11	0	40	15	18	5
Dyke2*^	17.504	78.869	15	103	−67	28	7	21	36	9	11	5
P3*^	12.649	77.496	17	143	−79	33	6	29	63	11	12	6
P10	12.470	77.320	4	18	67	136	8	50	96	11	13	6
P16*	13.509	76.582	16	77	−82	11	12	9	61	22	23	6
P26	14.075	77.280	4	71	−67	61	12	−1	39	16	20	6
P28	14.197	77.808	4	98	−68	180	7	16	38	10	12	6

Table 1. *(Continued.)*

Site	Lat (°N)	Long (°E)	N/n	D (°)	I (°)	k	α_{95} (°)	Plat. (°N)	Plong. (°E)	dp (°)	dm (°)	Ref
2	12.010	77.020	15	150	−84	150	3	22	71	6	6	7
3+4*	11.980	77.030	17	51	−76	26	7	−6	56	12	13	7
10*	11.890	76.950	5	30	70	196	6	42	100	8	9	7
12*	12.040	78.520	8	126	−71	55	8	30	47	12	13	7
16*	12.580	77.980	10	39	−81	63	6	−1	67	11	12	7
17*	12.630	78.070	5	186	−85	36	13	23	79	25	26	7
18*	13.490	76.580	8	105	−74	25	11	19	46	18	20	7
20*	14.310	76.630	5	73	−76	66	10	6	52	16	18	7
23	13.800	76.910	5	36	82	15	20	26	87	39	40	7
T3*	12.090	78.920	7	130	−74	481	3	30	52	5	5	8
D7*	12.080	77.890	6	106	−76	168	5	18	50	9	10	8
T4*	12.060	79.010	7	114	−68	92	10	25	40	13	16	8
T5*	12.050	79.030	6	307	−77	275	4	−3	99	7	8	8
T7*	12.050	79.080	7	92	−71	69	7	11	43	11	13	8
T8	12.110	79.100	3	121	−78	109	12	30	52	21	22	8
12*	12.660	77.500	6	135	−85	110	6	20	70	12	13	9
15*	12.650	77.420	6	73	−78	84	7	5	56	13	14	9
16*	12.660	77.420	5	60	−81	109	7	4	63	14	14	9
C	12.110	79.080	3	105	−75	280	7	18	50	12	13	10
D	12.100	78.916	3	169	−74	32	14	41	72	23	25	10
E*	12.110	79.070	6	115	−75	75	7	22	52	12	13	10
F	12.210	79.080	4	125	−75	58	9	27	53	15	16	10
Hol(1)	12.790	76.230	4	62	−78	916	3	2	57	5	6	11
Hol(2)	12.790	76.230	4	55	−79	218	6	0	59	11	12	11
i = A + B*^	14.190	77.640	5	57	−70	53	7	−7	47	10	12	12
ii^	14.180	77.760	3	72	−73	271	8	3	48	12	13	12
1*	12.900	78.200	7	170	−80	113	5	32	74	9	10	13
2*	12.900	78.200	6	88	−81	82	6	12	60	12	12	13
3*	12.900	78.200	9	127	−77	124	4	27	56	7	8	13
10	12.730	77.520	4	141	−75	50	10	34	57	17	18	14
Mean	***13.506***	***78.181***	***45/371***	***97***	***−80***	***56***	***3***	***15***	***58***	***6***	***6***	***Published***
Grand mean	***13.728***	***78.190***	***52/446***	***97***	***−79***	***55***	***3***	***15***	***57***	***5***	***6***	

Lat = site latitude, Long = site longitude, N/n: number of sites/number samples, D = mean declination, I= mean inclination, ^ = sites with geochronologic ages, α_{95} = cone of 95% confidence about the mean direction, k = precision parameter, Plat = pole latitude, Plong = pole longitude, dp = semi-axis of the confidence ellipse; dm = semi-axis perpendicular to dp, *= dykes used for calculation of the grand mean and satisfy acceptance criteria of N ≥ 5 and $\alpha_{95} \leq 15^\circ$ for the combined present study and published data. Mean directions and paleomagnetic pole for directions satisfying acceptance criteria are given in the table. References: 1. Valet *et al.* (2014), 2. Belica *et al.* (2014), 3. Radhakrishna *et al.* (2013), 4. Dash *et al.* (2013), 5. Kumar *et al.* (2012), 6. Piispa *et al.* (2011), 7. Halls *et al.* (2007), 8. Radhakrishna and Joseph (1996), 9. Dawson and Hargraves (1994), 10. Venkatesh *et al.* (1987), 11. Kumar (1985) (unpublished), 12. Kumar and Bhalla (1983), 13. Bhalla *et al.* (1980a), 14. Hasnain and Qureshy (1971).

4. Results

Two different paleomagnetic directions (components A and B) were isolated from the dataset of 29 sites in the eastern Dharwar craton. The present and published data with acceptance criteria of $N \geq 5$ and $\alpha_{95} \leq 15^\circ$ have been taken into consideration for the paleopole calculations.

4.1 *Component (A)*

Thirty-eight samples from seven sites representing four dykes (part of DGDS) have yielded a consistent paleomagnetic direction with easterly declination and steep upward inclination. This magnetic direction is named as component (A). Behaviour of alternating field (AF) and thermal demagnetization of representative samples is displayed in

Table 2. *Summary of paleomagnetic results from ~2367 Ma dykes with component (B).*

Site	Lat (°N)	Long (°E)	N/n	D (°)	I (°)	k	α_{95} (°)	Plat. (°N)	Plong. (°E)	dp (°)	dm (°)	Ref
Dyke 3												This study
DG 9*	15.490	77.670	10	49	4	21	11	40	179	6	11	
Dyke 5												
KA2*	18.126	78.685	11	50	−22	210	3	32	196	2	3	”
KA3*	18.126	78.685	13	56	−16	47	6	29	189	3	6	”
KA6*	18.177	78.784	9	56	12	39	8	34	174	4	9	”
KA7*	18.215	78.83	10	74	−17	33	9	12	182	5	9	”
KA11*	18.256	78.861	11	37	−36	18	11	37	213	7	13	”
KA14*	18.267	78.874	5	45	−6	26	15	41	190	8	15	”
KA15*	18.276	78.896	9	56	−32	38	9	25	199	5	10	”
KA21*	18.372	79.084	10	42	−11	20	11	42	195	6	11	”
Dyke 1	18.210	78.82	9	63	−6	9	16	24	181	8	16	3
Mean	**18.230**	**78.840**	**78**	**53**	**−17**	**14**	**5**	**32**	**192**	**3**	**5**	**”**
Dyke 6												
KA4*	18.167	78.833	12	44	−31	57	6	34	205	4	6	”
KA5*	18.181	78.853	10	49	−19	44	7	34	195	4	8	”
Mean	**16.860**	**78.255**	**22**	**47**	**−27**	**40**	**5**	**35**	**199**	**3**	**5**	**”**
Dyke 7												
HY 9	17.569	78.731	12	44	−19	17	11	39	197	6	11	”
Dyke 8												
IB2*	17.284	78.76	10	39	−30	34	8	39	207	5	9	”
IB3*	17.286	78.767	10	41	−20	45	7	41	200	4	8	”
Mean	**17.215**	**78.493**	**20**	**40**	**−25**	**36**	**6**	**40**	**203**	**3**	**6**	**”**
Dyke 9												
IB 12*	17.172	78.858	10	27	−17	45	7	52	210	4	8	”
Dyke 10												
IB4*	17.280	78.734	21	50	−25	65	4	32	198	2	4	”
IB14*	17.212	78.626	11	53	−7	28	9	34	185	4	9	”
IB18*	17.224	78.625	10	45	−6	173	4	42	189	2	4	”
Mean	**17.319**	**78.694**	**42**	**49**	**−16**	**31**	**4**	**35**	**192**	**2**	**4**	**”**
Dyke 11												
IB13*	17.191	78.846	10	53	−23	52	7	30	194	4	7	”
IB15*	17.117	78.717	9	48	−21	53	7	35	196	4	8	”
IB20*	17.105	78.692	20	38	−35	76	4	38	211	3	4	”
Mean	**17.235**	**78.682**	**39**	**44**	**−29**	**37**	**4**	**36**	**203**	**2**	**4**	**”**
Dyke 12												
DK 4*	16.654	78.915	11	35	−39	43	7	39	216	5	8	”
9 dykes mean	***17.082***	***78.571***	***22/244***	***41***	***−21***	***32***	***9***	***41***	***200***	***5***	***10***	***This study***
MG	14.259	78.06	4	63	−11	40	15	25	181	7	15	1
SC	14.092	77.77	3	63	14	69	15	28	167	8	15	1
P27m+Diii*												
P27m	14.196	77,808	14	66	−11	39	7	22	180	3	7	4
Dyke iii	14.194	77.806	12	64	−7	453	6	24	179	3	6	6
Mean	**14.195**	**77.807**	**26**	**64**	**−8**	**468**	**4**	**24**	**179**	**2**	**5**	**4 and 6**
29+D iv*												
29	14.181	77.729	4	67	−1	61	12	23	175	6	12	5
Dyke iv	14.187	77.735	9	57	−8	254	16	31	181	8	16	6
Mean	**14.184**	**77.732**	**13**	**60**	**−6**	**115**	**12**	**28**	**179**	**6**	**12**	**4 and 6**
P2*	16.830	77.59	7	48	−26	18	15	34	197	9	16	2
P3	16.820	77.71	7	36	7	12	18	53	185	9	18	2

Table 2. *(Continued.)*

Site	Lat (°N)	Long (°E)	N/n	D (°)	I (°)	k	α_{95} (°)	Plat. (°N)	Plong. (°E)	dp (°)	dm (°)	Ref
P13a	15.350	77.82	4	24	−4	23	20	60	202	10	20	2
P29	13.400	79.44	4	64	21	25	19	28	164	10	20	2
P70	14.160	78.61	4	61	−20	61	12	25	187	7	12	2
K.Nagar*	18.201	78.402	5	42	−19	39	6	40	199	3	6	5
Mean	***14.76***	***78.05***	***6/59***	***66***	***−16***	***28***	***12***	***21***	***183***	***6***	***12***	***Published***

Lat = site latitude, Long = site longitude, N/n: number of sites/number samples, D = mean declination, I= mean inclination, ^ = sites with Geochronologic ages, α_{95} = cone of 95% confidence about the mean direction, k = precision parameter, Plat = pole latitude, Plong = pole longitude, dp = semi-axis of the confidence ellipse; dm = semi-axis perpendicular to dp, *= Dykes used for calculation of the grand mean and satisfy acceptance criteria of N ≥ 5 and $\alpha_{95} \leq 15°$ for the combined present study and published data. Mean directions and paleomagnetic pole for directions satisfying acceptance criteria are given in the table. References: 1. Belica *et al.* (2014), 2. Radhakrishna *et al.* (2013), 3. Kumar *et al.* (2012), 4. Piispa *et al.* (2011), 5. Rao *et al.* (1990), 6. Kumar and Bhalla (1983).

figures 2(a–d) and 3(a–b), respectively. The characteristic remanent magnetization (ChRM) directions of these samples are isolated from high blocking temperatures. Results of thermal demagnetization technique shows that most of the samples intensity dropped to 5–10% between 560°C and 600°C. Isothermal remanent magnetization (IRM) studies were conducted on samples that carry component (A). The specimens are saturated in the range of 250–300 mT. Component (A) specimens coercivity of remanence values vary between 20 and 40 mT (figure 4) indicating that the remanence carrier could be magnetite. The mean ChRM (figure 5a) obtained for component (A) is D = 94°, I = −70° (k = 87, α_{95} = 10°). The paleopole calculated for this component (A) is P_{lat} = 16°N, P_{long} = 41°E (dp = 15°, dm = 17°) (figure 6a, table 1). Similar magnetic directions have been reported and confirmed as primary component of the ∼ 2367 Ma DGDS (Kumar and Bhalla 1983; Dash *et al.* 2013; Belica *et al.* 2014). Combining present and published data of all 52 sites (44 dykes) gives a mean D/I of 97°/–79° (k = 55, α_{95} = 3°) with a corresponding paleomagnetic pole at 15°N and 57°E (dp = 5°, dm = 6°; figure 6a, table 1).

4.2 *Component (B)*

One hundred and twenty samples from 22 sites representing eight dykes from the ∼ 2367 Ma DGDS are showing a paleomagnetic directions with north easterly declination and shallow upward and downward inclinations, which are named here as component (B). Behaviour of alternating field (AF) and thermal demagnetization of representative samples are displayed in figures 2 and 3, respectively. IRM studies were conducted on these samples that carry component (B). The specimens are saturated in the range from 300 and 350 mT. Component (B) specimens coercivity of remanence values vary between 30 and 50 mT (figure 4) indicating that the remanence carrier could be magnetite. The direction is consistent in all 22 sites (12 dykes) and the mean characteristic direction of component (B) are D = 41° and I = −21° (k = 32, α_{95} = 9°) with a corresponding paleomagnetic pole at 41°N and 200°E (dp = 5° and dm = 10°; figure 6b, table 2). Similar directions as that of component (B) is also observed on four more sites in the eastern Dharwar craton (EDC) and reported previously by several workers (Rao *et al.* 1990; Kumar *et al.* 2012, 2015; Dash *et al.* 2013; Radhakrishna *et al.* 2013). Combining present and earlier reported datasets representing 26 sites yields a grand mean of D/I is 46°/ − 20° (k = 19, α_{95} = 8°) with a corresponding paleomagnetic pole at 38°N and 196°E (dp = 4°, dm = 8°; figure 6b, table 2).

5. Discussion

Precise geochronology combined with consistent paleomagnetic results reported from dykes in the northern region and southern region of DGDS verifies the continuity of this swarm from north to south within the age range between 2365 and 2369 Ma (Halls *et al.* 2007; French and Heaman 2010; Demirer 2012). Whereas, from central part towards northern region (Dhone-Gooty), no paleomagnetic data is available so far. Karimnagar (northern part) and Bangalore dykes (southern part) of DGDS are mineralogically similar, only

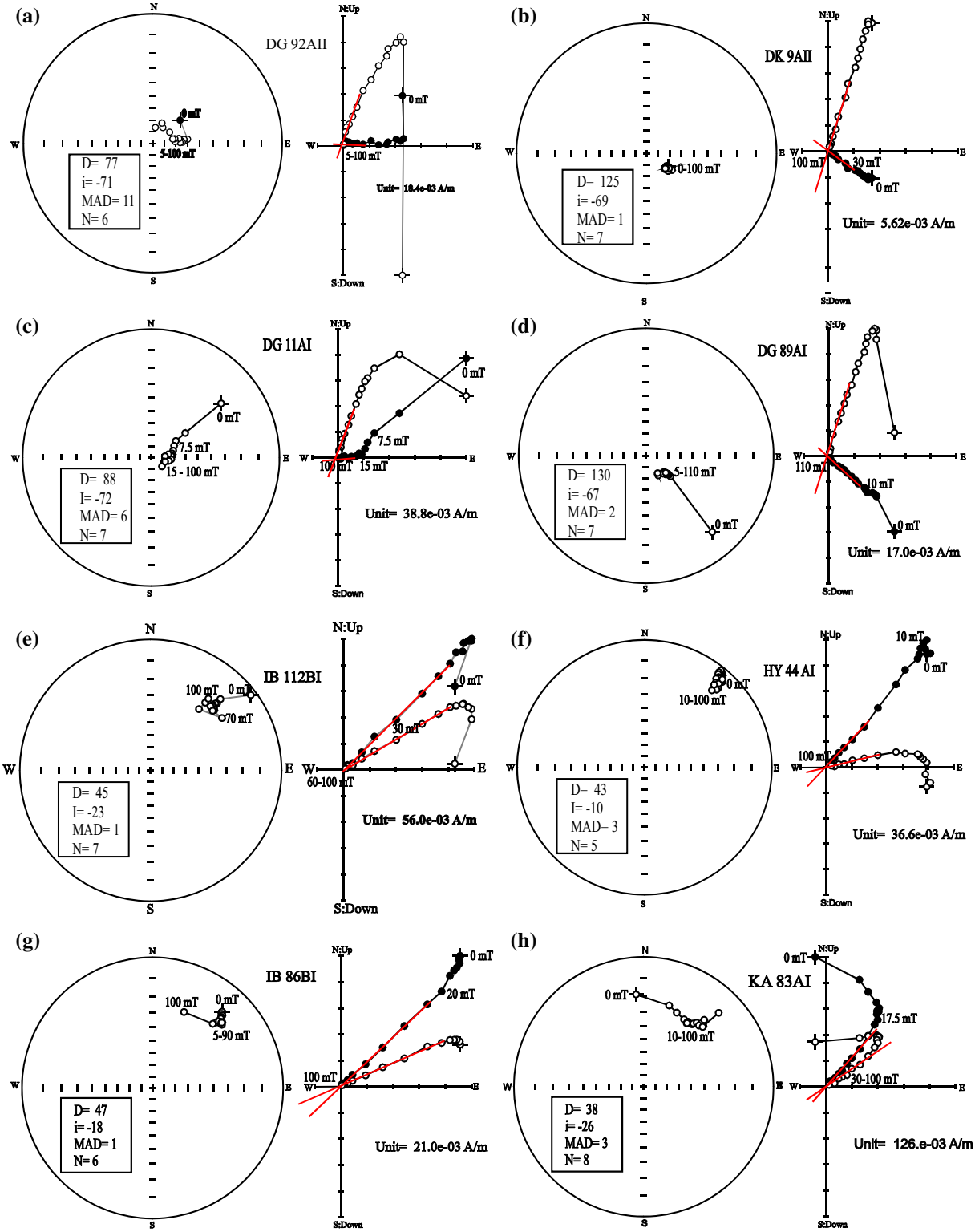


Figure 2. Equal area stereonet projections and zijderveld diagrams showing characteristic behaviour of AF demagnetization vectors for the samples that carry component (A) and component (B) are **(a–d)** and **(e–h)**, respectively. AF measurements are in mT. Open/closed circles in the stereoplots represent upward/downward directed vectors and open/closed circles in the zijderveld plots represent vertical/horizontal projections. Plotted using Remasoft 3.0, a plotting and analysis program (Chadima and Hrouda 2006).

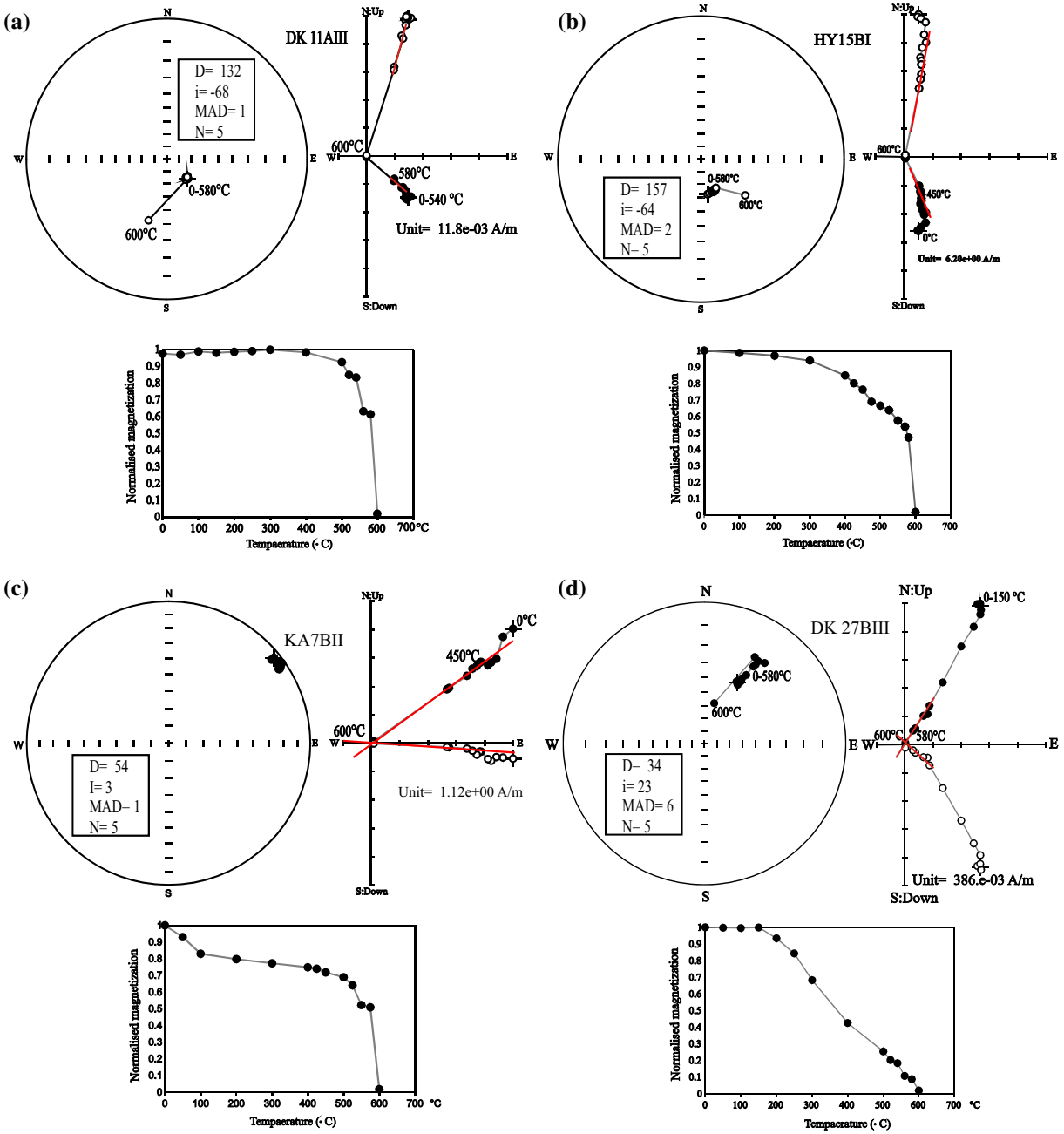


Figure 3. Equal area stereonet projections, zijderveld diagrams and intensity decay curves showing characteristic behaviour of thermal demagnetization vectors for the samples that carry component (A) and component (B) are **(a, b)** and **(c, d),** respectively. Thermal demagnetization steps are in °C. Open/closed circles in the stereoplots represent upward/downward directed vectors and open/closed circles in the zijderveld plots represent vertical/horizontal projections. Plotted using Remasoft 3.0, a plotting and analysis program (Chadima and Hrouda 2006).

differ in degree of alteration (Halls *et al.* 2007; Kumar *et al.* 2012; Srivastava *et al.* 2014c). Though the DGDS covers a large areal distribution, all dykes in this swarm are geochemically coherent and have similar primitive mantle-normalized trace element patterns and rare earth element characteristics (Halls *et al.* 2007; Kumar *et al.* 2012).

The obtained magnetization direction of component (A) is similar to the direction which has been reported earlier within DGDS from both northern and southern regions. The preservation of dual polarity and positive baked contact test (Kumar and Bhalla 1983; Halls *et al.* 2007; Dash *et al.* 2013; Radhakrishna *et al.* 2013; Belica *et al.* 2014) confirms the primary nature of magnetization. However, the present paleomagnetically yielded component (A) from the same trending dykes strengthens the database.

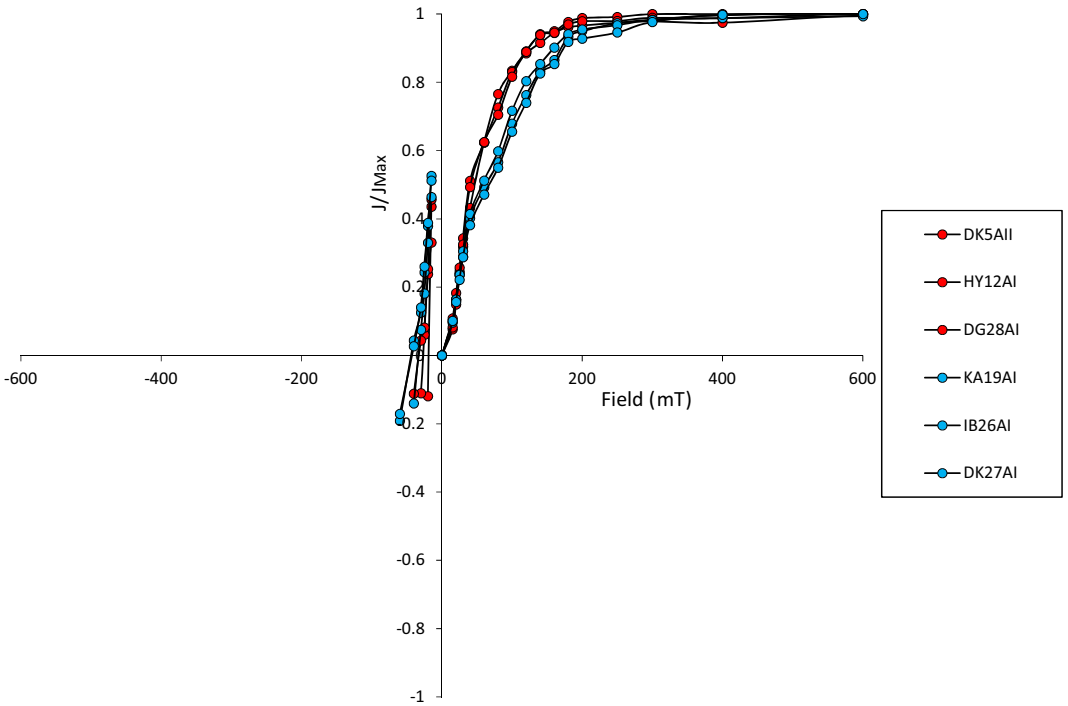


Figure 4. Isothermal remanent magnetization (IRM) acquisition curves for component (A) (solid red circles) and component (B) (solid blue circles) samples from DGDS dykes.

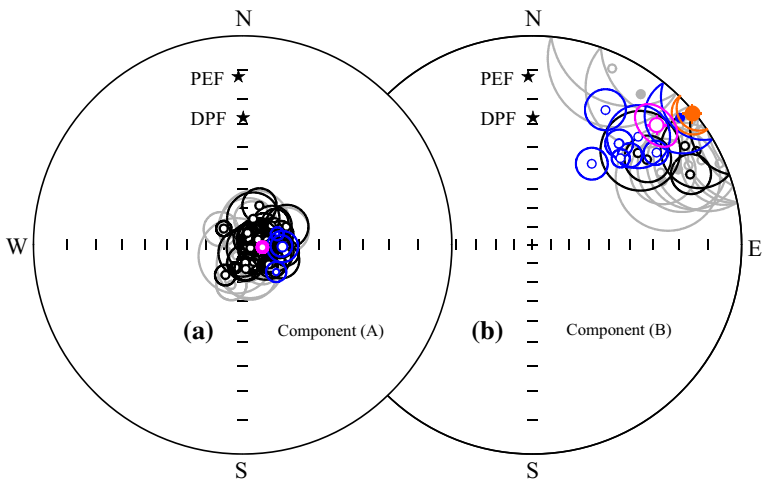


Figure 5. Stereoplots showing the site mean paleomagnetic data for present study and published sites (**a**) for component (A) and (**b**) for component (B). Present study (blue circles), published/accepted sites (black circles) and excluded sites (grey circles) with respect to component (A) and component (B) sites. Present study along with published sites grand mean direction of component (A) and (B) are represented as open pink circles. However, orange open circle in component (B) represents 20 Ma grand mean paleomagnetic direction. All the plotted data are shown in tables 1 and 2. Black stars represent DPF (Dipole Field) and PEF (Present Earth's Field) direction based on the 1995 IGRF.

The directions similar to component (B) are also observed earlier on DGDS and have been treated as secondary overprint (Halls *et al.* 2007; Kumar *et al.* 2012). This direction is considered as primary magnetization of 2082 Ma radiating dyke swarm of EDC (Kumar *et al.* 2015). Hence, we infer here that the source for component (B) is possibly the recently reported 2080 Ma spectacular radiating dyke swarm, which radiates beneath the Cuddapah basin with a trend from N134°W to N28°E and defines fan angle more than 160° (Kumar *et al.* 2015). The reason for different degree of overprint can be related to the population density of 2080 Ma dykes along DGDS. The component (B) has been

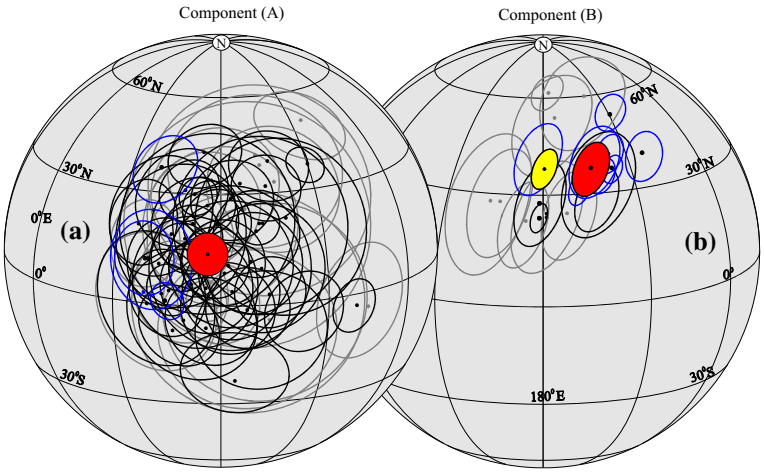


Figure 6. Orthogonal projection plot showing the paleopoles from Dharwar dyke swarm. (**a**) Paleopoles for component (A) and (**b**) paleopoles for component (B). Present study paleopoles are shown by blue circles, published/accepted sites paleopoles are shown by black circles and excluded sites paleopoles are shown by grey colour. Combined present study and published site means of component (A) and (B) represented as closed red circles. However, closed yellow circle represent the ∼ 2080 Ma grand mean (**b**). All the plotted data are shown in tables 1 and 2.

interpreted as an overprint acquired during the later tectonic event, accordant with the prior observations of ∼ 2080 Ma dyke swarm in the Dharwar craton.

5.1 *Reconstruction at ∼ 2.4 Ga*

In this section, we present a paleogeographic reconstruction among various cratons at ∼2.4 Ga. While performing this, we have tried to show cratonic blocks in a single configuration, so that the blocks can be brought closest together within the permissible constraints of the geology and paleomagnetic data. However, it should be noted that the paleolatitude and azimuthal orientation of each block are fixed by respective paleomagnetic data, but the paleolongitude and polarity are not defined (Buchan 2014).

Reliable key paleopoles at ∼2.4 Ga are limited to a few Archean cratons only. The mafic event, which is closest in terms of age to ∼2367 Ma DGDS is the E–W trending of Widgiemooltha dyke swarm (Yilgarn craton) with an emplacement age of 2418 ± 3 Ma (Nemchin and Pidgeon 1998) and 10 Ma younger Erayinia dykes (Yilgarn craton) with an emplacement age of 2401 ± 1 Ma (Pisarevsky *et al.* 2015). The initial Widgiemooltha paleomagnetic results of Evans (1968) were confirmed and strengthened by the more recent studies of Smirnov *et al.* (2013). In the paleogeographic reconstruction between Dharwar and Yilgran cratons, the potential link between Dharwar and Widgemooltha dykes based on the dyke patterns was proposed and it was concluded that a single magma plume is the source for these two swarms (Halls *et al.* 2007). Smirnov *et al.* (2013) have also proposed a possible link between Yilgarn craton and Zimbabwe craton based on the similarity in ages and paleopoles of Widgemooltha dykes and NNW trending Sebanaga Poort dyke (2408.3 ± 2.0 Ma; Soderlund *et al.* 2010). It would be interesting to see how the Dharwar, Yilgarn and Zimbabawe cratons fits together using paleopoles discussed above and listed in table 3. The Kola–Karelia craton is intruded by dense network of spectacular NW, E and NE trending dykes that are found from Russia to Finland. The geochronology work of Vuollo and Huhma (2005) has given at least five approximate age groups of mafic dykes from Kola–Karelia craton, i.e., 2.45, 2.32, 2.2, 2.1 and 1.98 Ga (Mertanen *et al.* 2006; Salminen *et al.* 2014). Since the 2.45–2.32 Ga dykes have similar paleopole (table 3), it is compelling to test how Kola–Karelia fit in the proposed paleogeographical reconstruction here.

We have attempted here a paleogeographical reconstruction using the free GPLATES software (http://www.gplates.org/) for coeval dyke events from Dharwar craton (India), Yilgarn craton (Australia), Kola–Karelia craton (Baltica shield) and Zimbabwe craton (Africa) at ∼2.4 Ga. All the paleopoles used for this reconstruction are listed in table 3. When plotted original data, paleoposition of Dharwar craton is located at steep latitude (68°) in the northern hemisphere. Using the freedom of paleolongitude, we have done the paleogeographic

Table 3. *Paleomagnetic poles for ~2400 Ma*

Name	Craton	Plat (°N)	Plong (°E)	A_{95} (°)	Age ref.	Paleomag ref.
Dharwar dykes (~ 2367 Ma)	Dharwar	15	57	3	Kumar *et al.* (2012); Demirer (2012); French and Heaman (2010); Halls *et al.* (2007)	Present study
Widgemooltha dykes (2418 ± 3 Ma)	Yilgarn	−10	159	8	Nemchin and Pidgeon (1998)	Smirnov *et al.* (2013); Evans (1968)
Erayinia dykes (2401 ± 1 Ma)	Yilgarn	−23	151	11	Pisarevsky *et al.* (2015)	Pisarevsky *et al.* (2015)
Karelian dykes (2450–2320 Ma)	Kola–Karelia	17	71	9	Vuollo and Huhma (2005)	Calculated by Pisarevsky *et al.* (2015) from Mertanen *et al.* (2006) and Salminen *et al.* (2014)
Sebanga dyke (2408.3 ± 2.0 Ma)	Zimbabwe	17	6	8	Soderlund *et al.* (2010)	Calculated by Smirnov *et al.* (2013) from Jones *et al.* (1975); Mushayandebvu *et al.* (1995)

A_{95} defines an oval of confidence within which there is a 95% probability that the true pole lies. Plat = pole latitude, Plong = pole longitude.

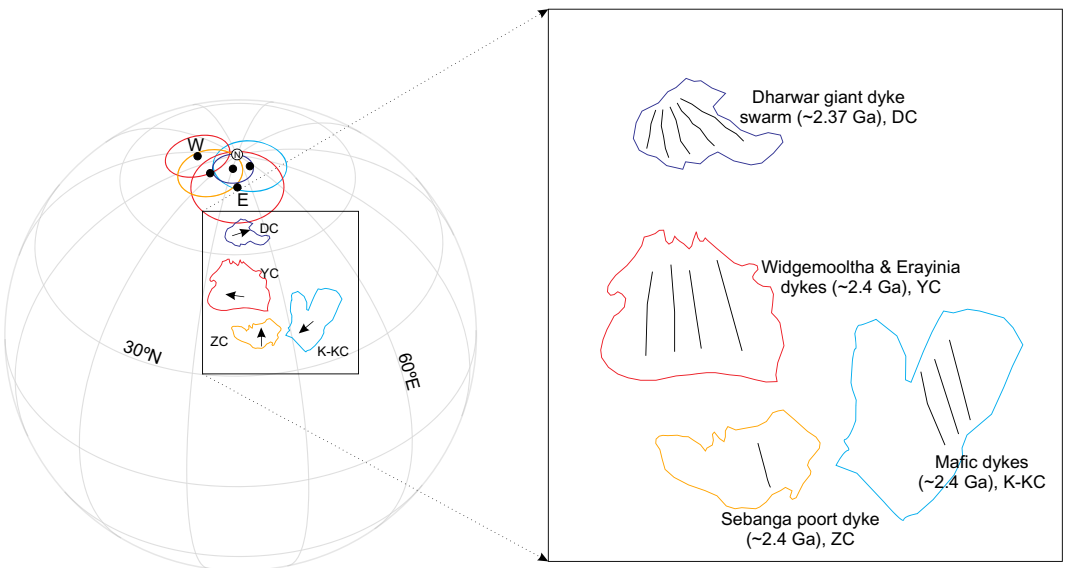


Figure 7. Orthogonal projection showing paleopositions of the Dharwar craton, Yilgarn craton, Zimbabwe craton and Kola–Karelia craton at ~2.4 Ga. All the cratons with their respective paleopoles are presented by following colour scheme, Dharwar by dark blue, Yilgarna by red, Zimbabwe by orange and Kola–Karelia by light. Coeval ~ 2.4 Ga dyke swarms are represented as black lines. 'DC' represents Dharwar craton, 'YC' represents Yilgarn craton, 'W' represents Widgemooltha dykes and 'E' represents Erayinia dykes 'ZC' represents Zimbabwe craton and K–KC represents Kola–Karelia craton. 'N' stands for the present North Pole.

reconstructions as shown in figure 7. The Euler rotation parameters for the paleogeographic reconstruction is compiled in table 4. In the present paleogeographic reconstruction (figure 7) the cratons are shown with their present known margins. The dykes are traced to emphasize similarity in the strike trends of all the dykes in their respective paleogeographic positions. This similarity in strike trends of the ~ 2.4 Ga dykes (figure 7) from Dharwar, Yilgarn, Zimbabwe and Kola–Karelia

Table 4. *Euler poles.*

Craton	λ (°N)	Φ (°E)	ω (degree)
Dharwar	49.80	80.10	−135.95
Yilgarn	24.15	−148.26	232.76
Kola-Karelia	−48.44	−133.11	221.661
Zimbabwe	−9.88	−85.22	−292.93

λ and Φ are the latitude and longitude of Euler poles respectively. ω is the Euler angle used for rotation. Euler angle (ω) is positive for counter-clockwise rotation and negative for clockwise rotation.

cratons in the reconstructed position suggests that the ∼ 2.4 Ga mafic magmatic event was fairly widespread. It should be noted here that the link between Yilgarn, Zimbabwe and Kola–Karelia cratons at ∼ 2.4 Ga has already been proposed and proved (Pisarevsky *et al.* 2015). We further propose that the ∼ 2.4 Ga mafic magmatic event has an extent of 50° latitude and 40° longitude based on our reconstruction (figure 7).

6. Conclusions

The paleomagnetic study of 12 NE–SW dykes from the present study strengthens the combined dataset for the Dharwar craton at ∼2367 Ma. The results of paleomagnetic components (A) and (B) are consistent with the results previously reported for the Dharwar dykes. We conclude here that the paleomagnetic directions of component (A) are primary and belong to the DGDS. The obtained paleopole (published and present study) for the component (A) of ∼2367 Ma DGDS is at 15°N and 57°E (dp = 5° and dm = 6°). The present study indicates that the ∼2080 Ma event may be strongly overprinted on the DGDS dykes in northern part of DGDS. The component (B) paleopole (published and present study) is at 38°N and 196°E (dp = 4° and dm = 8°).

The paleogeographical reconstruction for nearly coeval dyke events from Dharwar craton (India), Yilgarn craton (Australia), Kola-Karelia craton (Baltica shield) and Zimbabwe craton (Africa) at ca. 2.4 Ga is attempted in this study. This reconstruction shows the similarity in dyke trends and ages of key paleopoles from these cratons suggest that during ∼ 2.3 to ∼ 2.4 Ga Dharwar craton, Yilgarn craton, Zimbabwe craton and Kola–Karelia craton might have been spatially linked with each other.

Acknowledgements

We thank the Director, CSIR–NGRI, Hyderabad for the permission to publish this paper. We (NRB) would like to express our sincere gratitude to Dr Anil Kumar for his valuable advice and encouragement during this work. We also thank Prof. Joseph Meert for his valuable suggestions and constructive comments that helped to improve this paper to a large extent. We also thank the anonymous reviewer for his critical comments and suggestions. We also thank the handling editor Prof. Rajesh Srivastava.

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Corresponding editor: Rajesh Kumar Srivastava